

SHISHIR JAIMIN BHAVSAR

AI Developer & AI/ML Researcher | 3rd Year Diploma in CSE (SBMP) | Mumbai

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ABOUT ME

AI Developer & AI/ML Researcher and 3rd-year Diploma student in Computer Science Engineering at Shri Bhagubhai Mafatlal Polytechnic (SBMP), with a consistent distinction (82.6% board aggregate). Experienced in engineering high-performance AI systems (TARK: Rust CAS & WASM), automated cybersecurity architectures (KAVACH: AI SOAR & XDR), and mobile deep learning applications (Face Studio). Actively seeking an AI/ML or Software Engineering internship to build production-grade solutions.

EDUCATION

Diploma in Computer Science Engineering

2024 – 2027

Shri Bhagubhai Mafatlal Polytechnic College of Engineering

Current Standing: Distinction | 82.6% in SSC Board Exams

Secondary School Certificate (SSC)

2011 – 2024

G.E.S. English Medium School

TECHNICAL SKILLS

Languages	Python, Rust, TypeScript, JavaScript, Java, C#, C, HTML5, CSS3
AI / ML & Research	Deep Learning, PyTorch, Multi-Agent Systems, Generative AI & LLMs, HNSW Vector RAG, Computer Vision (CNNs), Symbolic Math (CAS)
Frameworks & Libs	FastAPI, React 18, Vite, WebAssembly (WASM), Android SDK, SymPy, KaTeX, Tailwind CSS
Systems & Cyber	SOAR & XDR Threat Defense, WebSocket Telemetry, MITRE ATT&CK, Linux, Penetration Testing (Cisco)
Databases & Tools	MySQL, Supabase, Git/GitHub, Docker, VS Code, Android Studio, AWS Cloud STTP
Soft Skills	Algorithmic Problem Solving, Technical Leadership, UI/UX Architecture, Rapid Prototyping

TECHNICAL PROJECTS

KAVACH – AI SOAR & XDR Platform

FastAPI / React / TypeScript

FastAPI | React | TypeScript | SOAR & XDR | AI Threat Defense | WebSockets

- Architected an enterprise-grade AI-powered SOAR & XDR cybersecurity platform delivering automated real-time threat detection and mitigation (<12ms response).
- Implemented FastAPI backend services with 12 database models, live WebSocket telemetry streaming, automated DAG response playbooks, and MITRE ATT&CK framework mapping.

TARK – Agentic Reasoning Platform

Rust / WASM / FastAPI / React / TS

Rust (CAS & Audio VAD) | WebAssembly | FastAPI | React 18 & Vite | HNSW Vector RAG | KaTeX

- Engineered an AI STEM reasoning platform featuring a native Rust Computer Algebra System (CAS) with AST parsing and symbolic differentiation to eliminate LLM mathematical hallucinations with zero-error verification.
- Built client-side 120 FPS WebAssembly physics simulations (RK4 double pendulum, N-body orbit), real-time Voice Activity Detection (VAD) oral exams, and sub-millisecond HNSW vector search over state curricula.

Face Studio – Android Application

DLAI 2026 | DAU & ISI Kolkata | github.com/facestudio4

- Built an AI-powered Android application as the capstone project for the DLAI 2026 (Deep Learning & Artificial Intelligence: Principles & Practice) workshop, jointly conducted by Dhirubhai Ambani University (DAU), Gandhinagar and Indian Statistical Institute (ISI), Kolkata. Actively in continued development.
- Achieves ~90% face detection accuracy (9/10 faces detected correctly) across standard lighting conditions, with response time of 1–2 seconds on a Redmi Note 12 (Android) — validated through real-device testing.
- Delivered a production APK as part of the workshop deliverables, demonstrating end-to-end mobile AI development.

To download the app — [click here](#)

RESEARCH & PUBLICATIONS

[Mechanistic Drivers of Sycophancy in Autoregressive Transformers](#)

DOI: [10.5281/zenodo.22929185](#)

[Mechanistic Interpretability](#) | [Causal Activation Patching](#) | [Direct Logit Attribution \(DLA\)](#) | [TransformerLens & Rust](#)

• Authored empirical research monograph on pythia-1.4b disproving the Parametric Retrieval Failure hypothesis (MLP recovery: 5.8%) via causal activation patching across 6 scientific domains (Geography, Astronomy, Physics, Chemistry, Biology, Math).

• Isolated dual-circuit suppression in L13–15 (Active Lie Boosters L13H6/L13H13 & Truth Gate L15H7); executed zero-shot surgical ablation reversing sycophancy (94.2% factual confidence preserved) with pure-Rust candle acceleration (~474k ops/s).

CERTIFICATIONS & WORKSHOPS

[DLAI 2026 – Deep Learning & Artificial Intelligence: Principles & Practice](#)

DAU, Gandhinagar & ISI Kolkata | 9–13 Mar 2026

Jointly organised by Dhirubhai Ambani University (DAU), Gandhinagar and Indian Statistical Institute (ISI), Kolkata · Backed by DST – Dept. of Science & Technology, Govt. of Gujarat and GUJCOST · Industry partners: MathWorks and GREENAI · Capstone: Face Studio Android App
The sessions spanned a wide spectrum of AI, including:

- Reinforcement Learning and its real-world applications
- Large Language Models and their evaluation
- Generative AI for Computer Vision
- Retrieval Augmented Generation (RAG)
- The evolution from Perceptrons to Transformers
- Image-based environmental intelligence and disaster prediction
- AI in smart farming and sensing systems

[AWS Cloud Practitioner Training – ISTE Approved STTP](#)

SVKM's SBMP College | 8–10 Jan 2026

Issued by SVKM's Shri Bhagubhai Mafatlal Polytechnic & College of Engineering · Approved by ISTE (Indian Society for Technical Education) · Covered AWS Cloud fundamentals, cloud infrastructure, and core practitioner-level concepts

[Cisco Networking Academy – Ethical Hacker](#)

Cisco Networking Academy | May 23, 2026

Certificate of Course Completion issued by Lynn Bloomer, Director, Cisco Networking Academy · Covered: Ethical hacking methodology, penetration testing, vulnerability scanning, social engineering, wired/wireless network exploits, application & cloud/mobile/IoT security, post-exploitation, and pentesting report creation

[Claude Code 101](#)

Anthropic | Jun 12, 2026

Certificate of Completion issued by Anthropic · Completed curriculum on agentic software development, terminal-based AI workflows (Claude Code CLI), test-driven agentic coding, and automated developer tooling

[Introduction to Generative AI](#)

IBM SkillsBuild | Jul 26, 2026

Certificate of Course Completion (ALM-COURSE_4058859) issued by IBM SkillsBuild · Completed 4 hours of coursework covering generative AI foundations, Large Language Models (LLMs), prompt engineering principles, and ethical AI deployment

[IGNITE IT 8.0 – 18-Hour Hackathon](#)

SVKM's SBMP College | 3–4 Sep 2026

Certificate of Participation issued by SVKM's Shri Bhagubhai Mafatlal Polytechnic & College of Engineering (IT Dept.) · Successfully completed an 18-hour intensive coding hackathon sprint focusing on collaborative development and rapid technical problem solving